

Table 1: NanoAOD 2024 datasets used in the analysis

Sample	NanoAOD dataset
Data (96 samples)	
Muon0_Run2024C	/Muon0/Run2024C-MINIV6NANOv15-v1/NANOAOD
Muon0_Run2024D	/Muon0/Run2024D-MINIV6NANOv15-v1/NANOAOD
Muon0_Run2024E	/Muon0/Run2024E-MINIV6NANOv15-v1/NANOAOD
Muon0_Run2024F	/Muon0/Run2024F-MINIV6NANOv15-v1/NANOAOD
Muon0_Run2024G	/Muon0/Run2024G-MINIV6NANOv15-v1/NANOAOD
Muon0_Run2024H	/Muon0/Run2024H-MINIV6NANOv15-v1/NANOAOD
Muon0_Run2024I_v1	/Muon0/Run2024I-MINIV6NANOv15-v1/NANOAOD
Muon0_Run2024I_v2	/Muon0/Run2024I-MINIV6NANOv15_v2-v1/NANOAOD
Muon1_Run2024C	/Muon1/Run2024C-MINIV6NANOv15-v1/NANOAOD
Muon1_Run2024D	/Muon1/Run2024D-MINIV6NANOv15-v1/NANOAOD
Muon1_Run2024E	/Muon1/Run2024E-MINIV6NANOv15-v1/NANOAOD
Muon1_Run2024F	/Muon1/Run2024F-MINIV6NANOv15-v1/NANOAOD
Muon1_Run2024G	/Muon1/Run2024G-MINIV6NANOv15-v2/NANOAOD
Muon1_Run2024H	/Muon1/Run2024H-MINIV6NANOv15-v2/NANOAOD
Muon1_Run2024I_v1	/Muon1/Run2024I-MINIV6NANOv15-v1/NANOAOD
Muon1_Run2024I_v2	/Muon1/Run2024I-MINIV6NANOv15_v2-v1/NANOAOD
MuonEG_Run2024C	/MuonEG/Run2024C-MINIV6NANOv15-v1/NANOAOD
MuonEG_Run2024D	/MuonEG/Run2024D-MINIV6NANOv15-v1/NANOAOD
MuonEG_Run2024E	/MuonEG/Run2024E-MINIV6NANOv15-v1/NANOAOD
MuonEG_Run2024F	/MuonEG/Run2024F-MINIV6NANOv15-v2/NANOAOD
MuonEG_Run2024G	/MuonEG/Run2024G-MINIV6NANOv15-v3/NANOAOD
MuonEG_Run2024H	/MuonEG/Run2024H-MINIV6NANOv15-v2/NANOAOD
MuonEG_Run2024I_v1	/MuonEG/Run2024I-MINIV6NANOv15-v2/NANOAOD
MuonEG_Run2024I_v2	/MuonEG/Run2024I-MINIV6NANOv15_v2-v2/NANOAOD
ParkingVBF0_Run2024C	/ParkingVBF0/Run2024C-MINIV6NANOv15-v1/NANOAOD
ParkingVBF0_Run2024D	/ParkingVBF0/Run2024D-MINIV6NANOv15-v1/NANOAOD
ParkingVBF0/Run2024E	/ParkingVBF0/Run2024E-MINIV6NANOv15-v1/NANOAOD
ParkingVBF0/Run2024F	/ParkingVBF0/Run2024F-MINIV6NANOv15-v3/NANOAOD
ParkingVBF0/Run2024G	/ParkingVBF0/Run2024G-MINIV6NANOv15-v3/NANOAOD
ParkingVBF0/Run2024H	/ParkingVBF0/Run2024H-MINIV6NANOv15-v2/NANOAOD
ParkingVBF0_Run2024I_v1	/ParkingVBF0/Run2024I-MINIV6NANOv15-v2/NANOAOD
ParkingVBF0_Run2024I_v2	/ParkingVBF0/Run2024I-MINIV6NANOv15_v2-v2/NANOAOD
ParkingVBF1_Run2024C	/ParkingVBF1/Run2024C-MINIV6NANOv15-v1/NANOAOD
ParkingVBF1_Run2024D	/ParkingVBF1/Run2024D-MINIV6NANOv15-v1/NANOAOD
ParkingVBF1_Run2024E	/ParkingVBF1/Run2024E-MINIV6NANOv15-v1/NANOAOD
ParkingVBF1_Run2024F	/ParkingVBF1/Run2024F-MINIV6NANOv15-v3/NANOAOD
ParkingVBF1_Run2024G	/ParkingVBF1/Run2024G-MINIV6NANOv15-v4/NANOAOD
ParkingVBF1_Run2024H	/ParkingVBF1/Run2024H-MINIV6NANOv15-v3/NANOAOD
ParkingVBF1_Run2024I_v1	/ParkingVBF1/Run2024I-MINIV6NANOv15-v3/NANOAOD
ParkingVBF1_Run2024I_v2	/ParkingVBF1/Run2024I-MINIV6NANOv15_v2-v3/NANOAOD
ParkingVBF2_Run2024C	/ParkingVBF2/Run2024C-MINIV6NANOv15-v1/NANOAOD
ParkingVBF2_Run2024D	/ParkingVBF2/Run2024D-MINIV6NANOv15-v2/NANOAOD
ParkingVBF2_Run2024E	/ParkingVBF2/Run2024E-MINIV6NANOv15-v1/NANOAOD
ParkingVBF2_Run2024F	/ParkingVBF2/Run2024F-MINIV6NANOv15-v3/NANOAOD
ParkingVBF2_Run2024G	/ParkingVBF2/Run2024G-MINIV6NANOv15-v3/NANOAOD
ParkingVBF2_Run2024H	/ParkingVBF2/Run2024H-MINIV6NANOv15-v3/NANOAOD
ParkingVBF2_Run2024I_v1	/ParkingVBF2/Run2024I-MINIV6NANOv15-v3/NANOAOD
ParkingVBF2_Run2024I_v2	/ParkingVBF2/Run2024I-MINIV6NANOv15_v2-v2/NANOAOD
ParkingVBF3_Run2024C	/ParkingVBF3/Run2024C-MINIV6NANOv15-v2/NANOAOD
ParkingVBF3_Run2024D	/ParkingVBF3/Run2024D-MINIV6NANOv15-v1/NANOAOD
ParkingVBF3_Run2024E	/ParkingVBF3/Run2024E-MINIV6NANOv15-v1/NANOAOD
ParkingVBF3_Run2024F	/ParkingVBF3/Run2024F-MINIV6NANOv15-v3/NANOAOD
ParkingVBF3_Run2024G	/ParkingVBF3/Run2024G-MINIV6NANOv15-v3/NANOAOD

Table 1: NanoAOD 2024 datasets used in the analysis (continued)

Sample	NanoAOD dataset
ParkingVBF3_Run2024H	/ParkingVBF3/Run2024H-MINIV6NANOv15-v3/NANOAOD
ParkingVBF3_Run2024I_v1	/ParkingVBF3/Run2024I-MINIV6NANOv15-v3/NANOAOD
ParkingVBF3_Run2024I_v2	/ParkingVBF3/Run2024I-MINIV6NANOv15_v2-v2/NANOAOD
ParkingVBF4_Run2024C	/ParkingVBF4/Run2024C-MINIV6NANOv15-v1/NANOAOD
ParkingVBF4_Run2024D	/ParkingVBF4/Run2024D-MINIV6NANOv15-v1/NANOAOD
ParkingVBF4_Run2024E	/ParkingVBF4/Run2024E-MINIV6NANOv15-v1/NANOAOD
ParkingVBF4_Run2024F	/ParkingVBF4/Run2024F-MINIV6NANOv15-v3/NANOAOD
ParkingVBF4_Run2024G	/ParkingVBF4/Run2024G-MINIV6NANOv15-v4/NANOAOD
ParkingVBF4_Run2024H	/ParkingVBF4/Run2024H-MINIV6NANOv15-v3/NANOAOD
ParkingVBF4_Run2024I_v1	/ParkingVBF4/Run2024I-MINIV6NANOv15-v3/NANOAOD
ParkingVBF4_Run2024I_v2	/ParkingVBF4/Run2024I-MINIV6NANOv15_v2-v2/NANOAOD
ParkingVBF5_Run2024C	/ParkingVBF5/Run2024C-MINIV6NANOv15-v1/NANOAOD
ParkingVBF5_Run2024D	/ParkingVBF5/Run2024D-MINIV6NANOv15-v1/NANOAOD
ParkingVBF5_Run2024E	/ParkingVBF5/Run2024E-MINIV6NANOv15-v1/NANOAOD
ParkingVBF5_Run2024F	/ParkingVBF5/Run2024F-MINIV6NANOv15-v3/NANOAOD
ParkingVBF5_Run2024G	/ParkingVBF5/Run2024G-MINIV6NANOv15-v3/NANOAOD
ParkingVBF5_Run2024H	/ParkingVBF5/Run2024H-MINIV6NANOv15-v3/NANOAOD
ParkingVBF5_Run2024I_v1	/ParkingVBF5/Run2024I-MINIV6NANOv15-v2/NANOAOD
ParkingVBF5_Run2024I_v2	/ParkingVBF5/Run2024I-MINIV6NANOv15_v2-v2/NANOAOD
ParkingVBF6_Run2024C	/ParkingVBF6/Run2024C-MINIV6NANOv15-v1/NANOAOD
ParkingVBF6_Run2024D	/ParkingVBF6/Run2024D-MINIV6NANOv15-v1/NANOAOD
ParkingVBF6_Run2024E	/ParkingVBF6/Run2024E-MINIV6NANOv15-v1/NANOAOD
ParkingVBF6_Run2024F	/ParkingVBF6/Run2024F-MINIV6NANOv15-v4/NANOAOD
ParkingVBF6_Run2024G	/ParkingVBF6/Run2024G-MINIV6NANOv15-v3/NANOAOD
ParkingVBF6_Run2024H	/ParkingVBF6/Run2024H-MINIV6NANOv15-v3/NANOAOD
ParkingVBF6_Run2024I_v1	/ParkingVBF6/Run2024I-MINIV6NANOv15-v2/NANOAOD
ParkingVBF6_Run2024I_v2	/ParkingVBF6/Run2024I-MINIV6NANOv15_v2-v2/NANOAOD
ParkingVBF7_Run2024C	/ParkingVBF7/Run2024C-MINIV6NANOv15-v1/NANOAOD
ParkingVBF7_Run2024D	/ParkingVBF7/Run2024D-MINIV6NANOv15-v1/NANOAOD
ParkingVBF7_Run2024E	/ParkingVBF7/Run2024E-MINIV6NANOv15-v1/NANOAOD
ParkingVBF7_Run2024F	/ParkingVBF7/Run2024F-MINIV6NANOv15-v3/NANOAOD
ParkingVBF7_Run2024G	/ParkingVBF7/Run2024G-MINIV6NANOv15-v3/NANOAOD
ParkingVBF7_Run2024H	/ParkingVBF7/Run2024H-MINIV6NANOv15-v4/NANOAOD
ParkingVBF7_Run2024I_v1	/ParkingVBF7/Run2024I-MINIV6NANOv15-v3/NANOAOD
ParkingVBF7_Run2024I_v2	/ParkingVBF7/Run2024I-MINIV6NANOv15_v2-v2/NANOAOD
Tau_Run2024C	/Tau/Run2024C-MINIV6NANOv15-v1/NANOAOD
Tau_Run2024D	/Tau/Run2024D-MINIV6NANOv15-v1/NANOAOD
Tau_Run2024E	/Tau/Run2024E-MINIV6NANOv15-v1/NANOAOD
Tau_Run2024F	/Tau/Run2024F-MINIV6NANOv15-v1/NANOAOD
Tau_Run2024G	/Tau/Run2024G-MINIV6NANOv15-v1/NANOAOD
Tau_Run2024H	/Tau/Run2024H-MINIV6NANOv15-v1/NANOAOD
Tau_Run2024I_v1	/Tau/Run2024I-MINIV6NANOv15-v1/NANOAOD
Tau_Run2024I_v2	/Tau/Run2024I-MINIV6NANOv15_v2-v1/NANOAOD
Simulation (177 samples)	
GluGluHto2Mu_M120	/GluGluH-Hto2Mu_Par-M-120_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
GluGluHto2Mu	/GluGluH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
GluGluHto2Mu_amcatnlo	/GluGluH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOADSIM
	/GluGluH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
GluGluHto2Mu_MinNLO	/GluGluH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powhegMINNLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
GluGluHto2Mu_tuneDown	/GluGluH-Hto2Mu_Par-M-125_TuneCP5Down_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
	/GluGluH-Hto2Mu_Par-M-125_TuneCP5Down_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOADSIM
GluGluHto2Mu_tuneUp	/GluGluH-Hto2Mu_Par-M-125_TuneCP5Up_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
GluGluHto2Mu_M130	/GluGluH-Hto2Mu_Par-M-130_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
	/GluGluH-Hto2Mu_Par-M-130_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOADSIM

Table 1: NanoAOD 2024 datasets used in the analysis (continued)

Sample	NanoAOD dataset
VBFHto2Mu_M125_powheg	/VBFH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM /VBFH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOADSIM
VBFHto2Mu_M120	/VBF-Hto2Mu_Par-M-120_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
VBFHto2Mu_m125_tuneCP5Up_amcatnlo	/VBF-Hto2Mu_Par-M-125_TuneCP5Up_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
VBFHto2Mu_m125_tuneCP5Down_amcatnlo	/VBF-Hto2Mu_Par-M-125_TuneCP5Down_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM /VBF-Hto2Mu_Par-M-125_TuneCP5Down_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOADSIM
VBFHto2Mu_M125_amcatnlo	/VBF-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM /VBF-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOADSIM
VBFHto2Mu_M130	/VBF-Hto2Mu_Par-M-130_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM /VBF-Hto2Mu_Par-M-130_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOADSIM
ggZH_Hto2Mu_ZtoAll_M125	/ggZH_HtoMuMu_ZToAll_M125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
VBFHto2Mu_m125_Flashsim	/VBFH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/fvaselli-VBFH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8-RunIII2024Summer24MiniAODv6-150X_mcRun3_2024_realistic_v2-v2-addHT1-c2e672ea58c2aebf452757dd48b38e22/USER /VBFH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/fvaselli-VBFH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8-RunIII2024Summer24MiniAODv6-150X_mcRun3_2024_realistic_v2_ext1-v2-addHT1-c2e672ea58c2aebf452757dd48b38e22/USER
ZH_Hto2Mu	/ZH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powhegMINLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOADSIM
WplusH_Hto2Mu	/WplusH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powhegMINLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOADSIM
WminusH_Hto2Mu	/WminusH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powhegMINLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOADSIM
TTH_Hto2Mu	/TTH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM /TTH-Hto2Mu_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOADSIM
DYto2Mu_M_50_0J_amcatnloFFFX	/DYto2Mu-2Jets_Bin-0J-MLL-50_TuneCP5_13p6TeV_amcatnloFFFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOADSIM
DYto2Mu_M_50_1J_amcatnloFFFX	/DYto2Mu-2Jets_Bin-1J-MLL-50_TuneCP5_13p6TeV_amcatnloFFFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v1/NANOADSIM
DYto2Mu_M_50_2J_amcatnloFFFX	/DYto2Mu-2Jets_Bin-2J-MLL-50_TuneCP5_13p6TeV_amcatnloFFFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOADSIM
DYto2Mu_M_10to50_amcatnloFFFX	/DYto2Mu-2Jets_Bin-MLL-10to50_TuneCP5_13p6TeV_amcatnloFFFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2Mu_M_50_amcatnloFFFX	/DYto2Mu-2Jets_Bin-MLL-50_TuneCP5_13p6TeV_amcatnloFFFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v6/NANOADSIM
DYto2Mu_MLL_105to160_amcatnloFFFX	/DYto2Mu-2Jets_Bin-MLL-105to160_TuneCP5_13p6TeV_amcatnloFFFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM /DYto2Mu-2Jets_Bin-MLL-105to160_TuneCP5_13p6TeV_amcatnloFFFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOADSIM
DYto2Mu_MLL_105to160_amcatnloFFFX_Fil_VBF	/DYto2Mu-2Jets_Bin-MLL-105to160_Fil-VBF_TuneCP5_13p6TeV_amcatnloFFFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2E_M_50_0J_amcatnloFFFX	/DYto2E-2Jets_Bin-0J-MLL-50_TuneCP5_13p6TeV_amcatnloFFFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2E_M_50_1J_amcatnloFFFX	/DYto2E-2Jets_Bin-1J-MLL-50_TuneCP5_13p6TeV_amcatnloFFFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOADSIM
DYto2E_M_50_2J_amcatnloFFFX	/DYto2E-2Jets_Bin-2J-MLL-50_TuneCP5_13p6TeV_amcatnloFFFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2E_M_10to50_amcatnloFFFX	/DYto2E-2Jets_Bin-MLL-10to50_TuneCP5_13p6TeV_amcatnloFFFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2E_M_50_amcatnloFFFX	/DYto2E-2Jets_Bin-MLL-50_TuneCP5_13p6TeV_amcatnloFFFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v4/NANOADSIM
DYto2Tau_M_50_0J_amcatnloFFFX	/DYto2Tau-2Jets_Bin-0J-MLL-50_TuneCP5_13p6TeV_amcatnloFFFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v4/NANOADSIM
DYto2Tau_M_50_1J_amcatnloFFFX	/DYto2Tau-2Jets_Bin-1J-MLL-50_TuneCP5_13p6TeV_amcatnloFFFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2Tau_M_50_2J_amcatnloFFFX	/DYto2Tau-2Jets_Bin-2J-MLL-50_TuneCP5_13p6TeV_amcatnloFFFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2Tau_M_10to50_amcatnloFFFX	/DYto2Tau-2Jets_Bin-MLL-10to50_TuneCP5_13p6TeV_amcatnloFFFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2Tau_M_50_amcatnloFFFX	/DYto2Tau-2Jets_Bin-MLL-50_TuneCP5_13p6TeV_amcatnloFFFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v7/NANOADSIM
DYto2Mu_MLL_10to50_powheg	/DYto2Mu_Bin-MLL-10to50_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2Mu_MLL_50to120_powheg	/DYto2Mu_Bin-MLL-50to120_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2Mu_MLL_120to200_powheg	/DYto2Mu_Bin-MLL-120to200_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2Mu_MLL_200to400_powheg	/DYto2Mu_Bin-MLL-200to400_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2Mu_MLL_400to800_powheg	/DYto2Mu_Bin-MLL-400to800_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2Mu_MLL_800to1500_powheg	/DYto2Mu_Bin-MLL-800to1500_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2Mu_MLL_1500to2500_powheg	/DYto2Mu_Bin-MLL-1500to2500_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2Mu_MLL_2500to4000_powheg	/DYto2Mu_Bin-MLL-2500to4000_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2Mu_MLL_4000to6000_powheg	/DYto2Mu_Bin-MLL-4000to6000_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2Mu_MLL_6000_powheg	/DYto2Mu_Bin-MLL-6000_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2Tau_MLL_10to50_powheg	/DYto2Tau_Bin-MLL-10to50_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2Tau_MLL_50to120_powheg	/DYto2Tau_Bin-MLL-50to120_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2Tau_MLL_120to200_powheg	/DYto2Tau_Bin-MLL-120to200_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2Tau_MLL_200to400_powheg	/DYto2Tau_Bin-MLL-200to400_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2Tau_MLL_400to800_powheg	/DYto2Tau_Bin-MLL-400to800_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2Tau_MLL_800to1500_powheg	/DYto2Tau_Bin-MLL-800to1500_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2Tau_MLL_1500to2500_powheg	/DYto2Tau_Bin-MLL-1500to2500_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2Tau_MLL_2500to4000_powheg	/DYto2Tau_Bin-MLL-2500to4000_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM

Continued on next page

Table 1: NanoAOD 2024 datasets used in the analysis (continued)

Sample	NanoAOD dataset
DYto2Tau_MLL_4000to6000_powheg	/DYto2Tau_Bin-MLL-4000to6000_TuneCP5_13p6TeV_powheg-pythia8/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2Tau_MLL_6000_powheg	/DYto2Tau_Bin-MLL-6000_TuneCP5_13p6TeV_powheg-pythia8/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2E_MLL_10to50_powheg	/DYto2E_Bin-MLL-10to50_TuneCP5_13p6TeV_powheg-pythia8/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2E_MLL_50to120_powheg	/DYto2E_Bin-MLL-50to120_TuneCP5_13p6TeV_powheg-pythia8/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2E_MLL_120to200_powheg	/DYto2E_Bin-MLL-120to200_TuneCP5_13p6TeV_powheg-pythia8/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2E_MLL_200to400_powheg	/DYto2E_Bin-MLL-200to400_TuneCP5_13p6TeV_powheg-pythia8/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2E_MLL_400to800_powheg	/DYto2E_Bin-MLL-400to800_TuneCP5_13p6TeV_powheg-pythia8/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2E_MLL_800to1500_powheg	/DYto2E_Bin-MLL-800to1500_TuneCP5_13p6TeV_powheg-pythia8/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2E_MLL_1500to2500_powheg	/DYto2E_Bin-MLL-1500to2500_TuneCP5_13p6TeV_powheg-pythia8/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2E_MLL_2500to4000_powheg	/DYto2E_Bin-MLL-2500to4000_TuneCP5_13p6TeV_powheg-pythia8/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2E_MLL_4000to6000_powheg	/DYto2E_Bin-MLL-4000to6000_TuneCP5_13p6TeV_powheg-pythia8/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2E_MLL_6000_powheg	/DYto2E_Bin-MLL-6000_TuneCP5_13p6TeV_powheg-pythia8/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2Mu_MLL_50to130_powheg_minnlo	/DYto2Mu_Bin-MLL-50to130_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photos/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2Mu_MLL_130to200_powheg_minnlo	/DYto2Mu_Bin-MLL-130to200_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photos/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2Mu_MLL_200to400_powheg_minnlo	/DYto2Mu_Bin-MLL-200to400_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photos/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2Mu_MLL_400to600_powheg_minnlo	/DYto2Mu_Bin-MLL-400to600_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photos/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2Mu_MLL_600to800_powheg_minnlo	/DYto2Mu_Bin-MLL-600to800_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photos/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2Mu_MLL_1000to1500_powheg_minnlo	/DYto2Mu_Bin-MLL-1000to1500_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photos/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2Mu_MLL_1500to2000_powheg_minnlo	/DYto2Mu_Bin-MLL-1500to2000_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photos/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2Mu_MLL_2000to4000_powheg_minnlo	/DYto2Mu_Bin-MLL-2000to4000_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photos/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2Mu_MLL_4000to6000_powheg_minnlo	/DYto2Mu_Bin-MLL-4000to6000_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photos/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2Mu_MLL_6000to13600_powheg_minnlo	/DYto2Mu_Bin-MLL-6000to13600_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photos/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
DYto2L_M_50_amcatnloFXXF_Flashsim	/DYto2L-2Jets_MLL-50_2J_TuneCP5_13p6TeV_amcatnloFXXF-pythia8/fvaselli-flashsim_DYto2L-2Jets_MLL-50_2J_TuneCP5_13p6TeV_amcatnloFXXF-pythia8-Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic-937767ad549695a10e40fd392f6c3633/USER
DYto2Mu_MLL_105to160_amcatnloFXXF_Flashsim	/DYto2Mu-2Jets_Bin-MLL-105to160_TuneCP5_13p6TeV_amcatnloFXXF-pythia8/fvaselli-DY2Mu_2Jets_MLL-105to160-RunII2024Summer24MiniAODv6-150X_mcRun3_2024_realistic_v2-packedTrackGenJetAK4-limit-TrkGJ_eta-to-tracker-66034083b387d860d393385d18a4f59d/USER
DYto2Mu_M_50_amcatnloFXXF_Flashsim	/DYto2Mu-2Jets_Bin-MLL-50_TuneCP5_13p6TeV_amcatnloFXXF-pythia8/fvaselli-SoftActivity_NanoAOD_Test_v1-ace5d3f4ddab40d8fc0f18e7a8dbab44/USER
EWK_2L2J_madgraph_herwig	/EWK-2L2J_Bin-MLL-50-MJJ-120_TuneCH3_13p6TeV_madgraph-herwig7/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
	/EWK-2L2J_Bin-MLL-50-MJJ-120_TuneCH3_13p6TeV_madgraph-herwig7/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOADSIM
EWK_2Mu2J_MLL_105to160_herwig	/EWK-2Mu2J_Bin-MLL-105to160_TuneCH3_13p6TeV_madgraph-herwig7/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
	/EWK-2Mu2J_Bin-MLL-105to160_TuneCH3_13p6TeV_madgraph-herwig7/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOADSIM
EWK_2Mu2J_MLL_105to160_pythia	/EWK-2Mu2J_Bin-M2Mu-105to160-M2J-120_TuneCP5_13p6TeV_madgraph-pythia8/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
	/EWK-2Mu2J_Bin-M2Mu-105to160-M2J-120_TuneCP5_13p6TeV_madgraph-pythia8/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOADSIM
TTLNu-EWK_amcatnlo	/TTLNu-EWK_TuneCP5_13p6TeV_amcatnlo-pythia8/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
WmWmJJ_EWK	/WmWmJJ-EWK_TuneCP5_13p6TeV-powheg-pythia8/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
EWK_2Mu2J_MLL_105to160_pythia_Flashsim	/EWK-2Mu2J_Bin-M2Mu-105to160-M2J-120_TuneCP5_13p6TeV_madgraph-pythia8/fvaselli-EWK-2Mu2J_Bin-M2Mu-105to160-M2J-120-RunII2024Summer24MiniAODv6-150X_mcRun3_2024_realistic_v2-addHT1-c2e672ea58c2aebf452757dd48b38e22/USER
ggZH_Hto2B_Zto2Q	/GluGluZH-Zto2Q-Hto2B_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
ggZH_Hto2B_Zto2L	/GluGluZH-Zto2L-Hto2B_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
GluGluHto2B_M125	/GluGluH-Hto2B_Par-M-125_TuneCP5_13p6TeV_powhegMINLO-pythia8/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
	/GluGluH-Hto2B_Par-M-125_TuneCP5_13p6TeV_powhegMINLO-pythia8/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOADSIM
GluGluHto2Tau_M125	/GluGluHto2Tau_Par-M-125_TuneCP5_13p6TeV_powhegMINNLO-pythia8/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
GluGluHto2Tau_UnCorrelatedDecay_Filtered	/GluGluH-Hto2TauUnCorrelatedDecay_Fil-TauFilter_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
GluGluHto2Tau_UnCorrelatedDecay_UnFiltered	/GluGluH-Hto2TauUnCorrelatedDecay_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
GluGluHto2Wto2L2Nu_M125	/GluGluHto2Wto2L2Nu_Par-M-125_TuneCP5_13p6TeV_powheg-jhugen-pythia8/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
TTHto2B_M125	/TTH-Hto2B_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
TTHtoNon2B_M125	/TTH-HtoNon2B_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
TTZH_ZHto4B	/TTZH-ZHto4B_TuneCP5_13p6TeV_madgraph-pythia8/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
VBFHto2B_M125	/VBFH-Hto2B_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
	/VBFH-Hto2B_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2_ext1-v2/NANOADSIM
VBFHto2Tau_M125	/VBFH-Hto2Tau_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
VBFHto2Tau_UnCorrelatedDecay_UnFiltered	/VBFH-Hto2TauUnCorrelatedDecay_Fil-TauFilter_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
VBFHto2Tau_UnCorrelatedDecay_Filtered	/VBFH-Hto2TauUnCorrelatedDecay_Par-M-125_TuneCP5_13p6TeV_powheg-pythia8/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
VBFHto2Wto2L2Nu_M125	/VBFHto2Wto2L2Nu_Par-M-125_TuneCP5_13p6TeV_powheg-jhugen-pythia8/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
WminusH_Hto2B_WtoLnu	/WminusH-WtoLnu-Hto2B_Par-M-125_TuneCP5_13p6TeV_powhegMINLO-pythia8/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
WplusH_Hto2B_WtoLnu	/WplusH-WtoLnu-Hto2B_Par-M-125_TuneCP5_13p6TeV_powhegMINLO-pythia8/RunII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM

Continued on next page

Table 1: NanoAOD 2024 datasets used in the analysis (continued)

Sample	NanoAOD dataset
WminusHto2Tau_UncorrelatedDecay_Filtered	/WminusH-Hto2TauUncorrelatedDecay_Fil-TauFilter_Par-M-125_TuneCP5_13p6TeV_powhegMINNLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
WminusHto2Tau_UncorrelatedDecay_UnFiltered	/WminusH-Hto2TauUncorrelatedDecay_Par-M-125_TuneCP5_13p6TeV_powhegMINNLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
WplusHto2Tau_UncorrelatedDecay_Filtered	/WplusH-Hto2TauUncorrelatedDecay_Fil-TauFilter_Par-M-125_TuneCP5_13p6TeV_powhegMINNLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
WplusHto2Tau_UncorrelatedDecay_UnFiltered	/WplusH-Hto2TauUncorrelatedDecay_Par-M-125_TuneCP5_13p6TeV_powhegMINNLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
ZH_Hto2B_Zto2L	/ZH-Zto2L-Hto2B_Par-M-125_TuneCP5_13p6TeV_powhegMINLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
ZH_Hto2B_Zto2Q	/ZH-Zto2Q-Hto2B_Par-M-125_TuneCP5_13p6TeV_powhegMINLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
TbarBtoLminusNuB_s_channel_4FS	/TbarBtoLminusNuB-s-channel-4FS_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
TBbartoLplusNuBbar_s_channel_4FS	/TBbartoLplusNuBbar-s-channel-4FS_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
TbarBQtoLnu_t_channel_4FS	/TbarBQtoLnu-t-channel-4FS_TuneCP5_13p6TeV_powheg-madspin-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
TbarBQto2Q_t_channel_4FS	/TbarBQto2Q-t-channel-4FS_TuneCP5_13p6TeV_powheg-madspin-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
TBbarQtoLnu_t_channel_4FS	/TBbarQtoLnu-t-channel-4FS_TuneCP5_13p6TeV_powheg-madspin-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
TBbarQto2Q_t_channel_4FS	/TBbarQto2Q-t-channel-4FS_TuneCP5_13p6TeV_powheg-madspin-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
TbarWplusto2L2Nu	/TbarWplusto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
TbarWplusto4Q	/TbarWplusto4Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
TbarWplustoLnu2Q	/TbarWplustoLnu2Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
TWminusto2L2Nu	/TWminusto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
TWminusto4Q	/TWminusto4Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
TWminustoLnu2Q	/TWminustoLnu2Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
TTto2L2Nu	/TTto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOADSIM
TTto4Q	/TTto4Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
TTtoLnu2Q	/TTtoLnu2Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
TTWH	/TTWH_TuneCP5_13p6TeV_madgraph-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
TTWW	/TTWW_TuneCP5_13p6TeV_madgraph-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
TTWZ	/TTWZ_TuneCP5_13p6TeV_madgraph-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
TTZZ_ZZto4B	/TTZZ-ZZto4B_TuneCP5_13p6TeV_madgraph-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOADSIM/
TTZ_Zto2Q	/TTZ-ZtoQQ-1Jets_TuneCP5_13p6TeV_amcatnloFXFXold-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
TTto2L2Nu_Flashsim	/TTto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/fvaselli-TTto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8-RunIII2024Summer24MiniAODv6-150X_mcRun3_2024_realistic_v2-addHT1-c2e672ea58c2aebf452757dd48b38e22/USER
TT_Flashsim	/TT_TuneCH3_13TeV_powheg-herwig7/fvaselli-flashsim_v9-f5e5c27141f6cbae6ce3c7d60feb1683/USER
WtoLnu_1J_madgraphMLM	/WtoLnu-4Jets_Bin-1J_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
WtoLnu_2J_amcatnloFXFX	/WtoLnu-4Jets_Bin-2J_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
WtoLnu_2J_madgraphMLM	/WtoLnu-4Jets_Bin-2J_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
WtoLnu_3J_madgraphMLM	/WtoLnu-4Jets_Bin-3J_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
WtoLnu_4J_madgraphMLM	/WtoLnu-4Jets_Bin-4J_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
WtoLnu_PTLNu_40to100_1J_amcatnloFXFX	/WtoLnu-2Jets_Bin-1J-PTLnu-40to100_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOADSIM
WtoLnu_PTLNu_40to100_2J_amcatnloFXFX	/WtoLnu-2Jets_Bin-2J-PTLnu-40to100_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOADSIM
WtoLnu_PTLNu_100to200_1J_amcatnloFXFX	/WtoLnu-2Jets_Bin-1J-PTLnu-100to200_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOADSIM
WtoLnu_PTLNu_100to200_2J_amcatnloFXFX	/WtoLnu-2Jets_Bin-2J-PTLnu-100to200_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOADSIM
WtoLnu_PTLNu_200to400_1J_amcatnloFXFX	/WtoLnu-2Jets_Bin-1J-PTLnu-200to400_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
WtoLnu_PTLNu_200to400_2J_amcatnloFXFX	/WtoLnu-2Jets_Bin-2J-PTLnu-200to400_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
WtoLnu_PTLNu_400to600_1J_amcatnloFXFX	/WtoLnu-2Jets_Bin-1J-PTLnu-400to600_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
WtoLnu_PTLNu_400to600_2J_amcatnloFXFX	/WtoLnu-2Jets_Bin-2J-PTLnu-400to600_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
WtoLnu_PTLNu_600_1J_amcatnloFXFX	/WtoLnu-2Jets_Bin-1J-PTLnu-600_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
WtoLnu_PTLNu_600_2J_amcatnloFXFX	/WtoLnu-2Jets_Bin-2J-PTLnu-600_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
WtoENu_amcatnloFXFX	/WtoENu-2Jets_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOADSIM
WtoMuNu_amcatnloFXFX	/WtoMuNu-2Jets_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOADSIM
WtoTauNu_amcatnloFXFX	/WtoTauNu-2Jets_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOADSIM
WW	/WW_TuneCP5_13p6TeV_pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
WZ	/WZ_TuneCP5_13p6TeV_pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
ZZ	/ZZ_TuneCP5_13p6TeV_pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
WWto2L2Nu_powheg	/WWto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
WWto4Q_powheg	/WWto4Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
WWtoLnu2Q_powheg	/WWtoLnu2Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
WZto2L2Q_powheg	/WZto2L2Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM

Continued on next page

Table 1: NanoAOD 2024 datasets used in the analysis (continued)

Sample	NanoAOD dataset
WZto3LNu_powheg	/WZto3LNU_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
WZtoLNU2Q_powheg	/WZtoLNU2Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
ZZto2L2Nu_powheg	/ZZto2L2NU_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
ZZto2L2Q_powheg	/ZZto2L2Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
ZZto2Nu2Q_powheg	/ZZto2NU2Q_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
ZZto4L_powheg	/ZZto4L_TuneCP5_13p6TeV_powheg-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
WWW_4F	/WWW-4F_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
WWZ_4F	/WWZ-4F_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
WZZ	/WZZ-5F_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
ZZZ	/ZZZ-5F_TuneCP5_13p6TeV_amcatnlo-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
ZHto2Tau_UncorrelatedDecay_Filtered	/ZH-Hto2TauUncorrelatedDecay_Fil-TauFilter_Par-M-125_TuneCP5_13p6TeV_powhegMINNLO-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v6/NANOADSIM
Zto2Nu_HT_100to200	/Zto2Nu-4Jets_Bin-HT-100to200_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOADSIM
Zto2Nu_HT_200to400	/Zto2Nu-4Jets_Bin-HT-200to400_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOADSIM
Zto2Nu_HT_400to800	/Zto2Nu-4Jets_Bin-HT-400to800_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
Zto2Nu_HT_800to1500	/Zto2Nu-4Jets_Bin-HT-800to1500_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
Zto2Nu_HT_1500to2500	/Zto2Nu-4Jets_Bin-HT-1500to2500_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
Zto2Nu_HT_2500toInf	/Zto2Nu-4Jets_Bin-HT-2500_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
Zto2Q_HT_100to400	/Zto2Q-4Jets_Bin-HT-100to400_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
Zto2Q_HT_400to800	/Zto2Q-4Jets_Bin-HT-400to800_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
Zto2Q_HT_800to1500	/Zto2Q-4Jets_Bin-HT-800to1500_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
Zto2Q_HT_1500to2500	/Zto2Q-4Jets_Bin-HT-1500to2500_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v2/NANOADSIM
Zto2Q_HT_2500toInf	/Zto2Q-4Jets_Bin-HT-2500_TuneCP5_13p6TeV_madgraphMLM-pythia8/RunIII2024Summer24NanoAODv15-150X_mcRun3_2024_realistic_v2-v3/NANOADSIM